Comparative Evaluation of ImageNet-Pretrained ResNet-18 and DeiT-Tiny for Pediatric Pneumonia Classification

Full Fine-Tuning on PneumoniaMNIST+ 224

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Abstract. This study compares an ImageNet-pretrained convolutional neural network (ResNet-18) and Vision Transformer (DeiT-Tiny) for binary classification of pediatric chest radiographs. Both models were fully fine-tuned using the same preprocessing, AdamW settings, five-epoch budget, and three predefined training seeds on PneumoniaMNIST+ 224. The best checkpoint of each run was selected using validation macro-F1, then evaluated on the official held-out test set of 624 images. Mean test macro-F1 was **0.9114 ± 0.0176** for ResNet-18 and **0.9121 ± 0.0119** for DeiT-Tiny. Both models had very high sensitivity (0.9974 on average) but lower specificity (about 0.79), indicating more errors among Normal images. Training times were similar, and no model performed consistently better across all seeds. These results support comparable performance for the *evaluated pipelines*, not general architectural or clinical equivalence.

Keywords: biomedical imaging; transfer learning; ResNet-18; DeiT-Tiny; PneumoniaMNIST+.

1 Introduction

Chest radiograph classification is a useful setting for studying how visual representations learned from natural images transfer to biomedical images. In this work, the question is: *How do ImageNet-pretrained ResNet-18 and DeiT-Tiny compare in pediatric pneumonia classification performance and training time under a shared full fine-tuning protocol?*

ResNet-18 uses residual convolutional blocks (He et al., 2016); DeiT-Tiny processes image patch tokens using self-attention (Touvron et al., 2021). Both start from ImageNet-1K pretrained weights (Russakovsky et al., 2015). The purpose here is not to propose a new architecture or a clinical diagnostic system. It is to carry out a reproducible, small-scale comparison and report both average outcomes and run-to-run variability.

2 Dataset and Preprocessing

The official PneumoniaMNIST+ 224 collection provides 5,856 grayscale images derived from a pediatric chest X-ray dataset (Kermany et al., 2018; Yang et al., 2023; MedMNIST Project, 2024). We used the supplied splits, without resampling or creating additional partitions. Class 0 denotes Normal and class 1 denotes Pneumonia.

Table 1: Official dataset sizes and label distribution, reproduced from notebook outputs.

Split	Normal	Pneumonia	Total	Pneumonia (%)
Training	1,214	3,494	4,708	74.21
Validation	135	389	524	74.24
Test	234	390	624	62.50

The original 224×224 grayscale images were converted to three identical channels, transformed to float tensors, and normalized using ImageNet channel means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. The same deterministic preprocessing was applied to all splits; no random data augmentation, class reweighting, or model-specific input preprocessing was used.

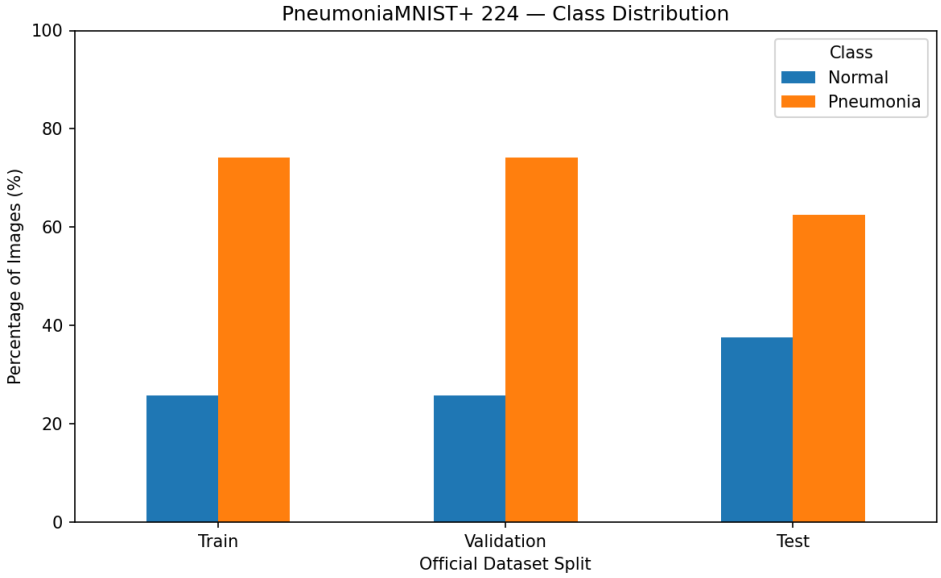


Figure 1: Class distribution across official splits. Pneumonia represents approximately 74.2% of training and validation data, versus 62.5% of the test data.

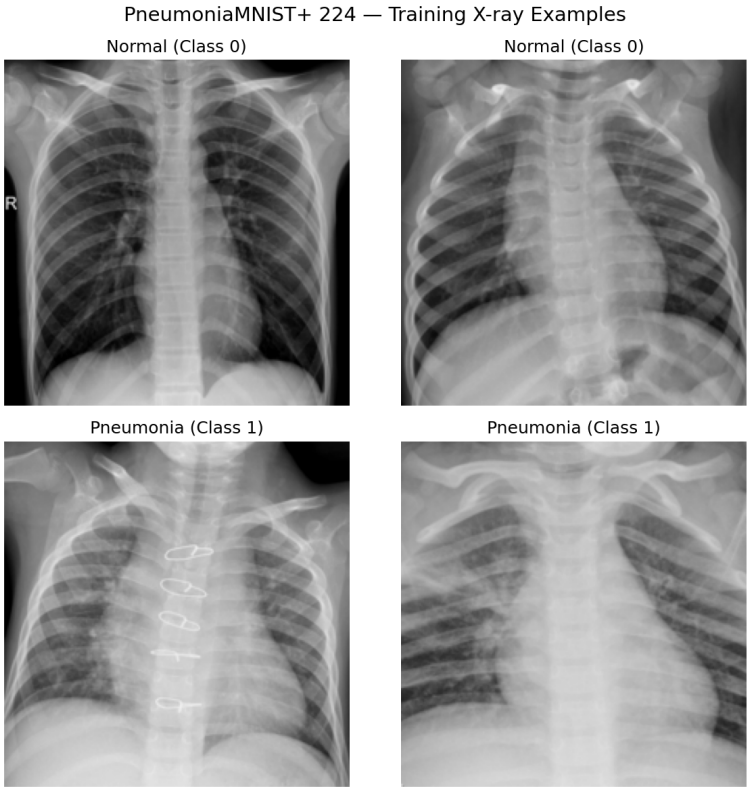


Figure 2: Four illustrative training images: two Normal and two Pneumonia, using the supplied dataset labels. These examples are not independently interpreted radiographs.

3 Models and Experimental Protocol

3.1 Pretrained models

We used the `timm` library (Hugging Face, 2026) to load `resnet18.tv_in1k` (**11,177,538** total/trainable parameters) and `deit_tiny_patch16_224.fb_in1k` (**5,524,802** total/trainable parameters). The classifier was configured for two output logits in both cases. DeiT-Tiny uses 16×16 patches: 196 patch tokens plus one CLS token, with embedding dimension 192, 12 encoder blocks, and three attention heads per block. The implemented model is the non-distilled DeiT-Tiny variant; no distillation procedure was run in this study. All model parameters were trainable.

3.2 Training and checkpoint selection

Each model was restarted from its corresponding ImageNet-pretrained weights for each seed (42, 43, 44). A separate optimizer and seeded training DataLoader were created for every run. All six runs used the same configuration:

Table 2: Fixed full fine-tuning and evaluation protocol.

Input size	224×224, three channels	**Batch size**	16
Optimizer	AdamW	**Learning rate**	2×10^{-5}
Weight decay	0.01	**Loss**	Cross-entropy (unweighted)
Epochs/run	5 (fixed)	**Seeds**	42, 43, 44
Model selection	Highest validation macro-F1	**Test set**	Final evaluation only

For each run, training and validation losses and validation macro-F1 were recorded after every epoch. The checkpoint achieving the highest validation macro-F1 was saved; exact ties retained the earlier epoch. No learning-rate search, model-specific hyperparameter tuning, early stopping, or new training was performed after inspecting the final test results.

Training used PyTorch on Apple Silicon MPS, as recorded in the notebook (Python 3.13.5, PyTorch 2.14.1, `timm` 1.0.31, `medmnist` 3.0.2). Runtime is the notebook's elapsed training/validation/checkpoint time per five-epoch run, *not* inference latency or a formal device benchmark.

3.3 Evaluation metrics

The primary outcome was **macro-F1**, the average F1 across Normal and Pneumonia, which gives each class equal weight. We also measured accuracy, sensitivity (Pneumonia recall), specificity (Normal recall), and ROC-AUC (from predicted Pneumonia probabilities). With TN, FP, FN, TP computed using class order [Normal, Pneumonia],

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \qquad \text{Specificity} = \frac{TN}{TN + FP}. \tag{1}$$

Each checkpoint was evaluated on the same held-out 624-image test set. Per-model results are shown as mean $\pm$ *sample* standard deviation over three training seeds ($n = 3$, `ddof=1`); these values are not confidence intervals for new patients.

4 Experimental Results

4.1 Per-run validation selection and test outcomes

Table 3 combines the selected checkpoint epoch, validation macro-F1, held-out test macro-F1, false-positive and false-negative counts, and recorded runtime for each model–seed run. All confusion matrices contain 234 Normal and 390 Pneumonia images.

Table 3: Run-level results from saved validation/test CSV files and confusion matrices. Runtime is measured over five epochs.

Model	Seed	Best epoch	Val. F1	Test F1	FP	FN	Time (min)
ResNet-18	42	4	0.9875	0.8910	59	1	1.19
ResNet-18	43	4	0.9900	0.9216	43	1	1.21
ResNet-18	44	5	0.9876	0.9216	43	1	1.72
DeiT-Tiny	42	5	0.9826	0.9255	40	2	1.38
DeiT-Tiny	43	5	0.9824	0.9026	53	1	1.32
DeiT-Tiny	44	4	0.9776	0.9082	51	0	1.37

ResNet-18 achieved higher mean best-validation macro-F1 (0.9884 ± 0.0014) than DeiT-Tiny (0.9809 ± 0.0028). The corresponding means for training runtime were 1.37 and 1.36 minutes/run, respectively. The measurements are close; this experiment did not demonstrate a clear runtime advantage.

4.2 Final held-out test performance

Table 4: Held-out test metrics: mean ± sample SD across three seeds. Macro-F1 is the primary comparison metric.

Metric	ResNet-18	DeiT-Tiny
Macro-F1 (primary)	0.9114 ± 0.0176	0.9121 ± 0.0119
Accuracy	0.9209 ± 0.0148	0.9215 ± 0.0100
Sensitivity	0.9974 ± 0.0000	0.9974 ± 0.0026
Specificity	0.7934 ± 0.0395	0.7949 ± 0.0299
ROC-AUC	0.9907 ± 0.0027	0.9900 ± 0.0005

Mean test macro-F1 differed by just 0.0007 (0.07 percentage points). The preferred model depended on seed: DeiT-Tiny scored higher for seed 42, while ResNet-18 scored higher for seeds 43 and 44. This is evidence of *comparable performance under the tested conditions*, not proof of statistical equivalence or superiority.

4.3 Learning behavior

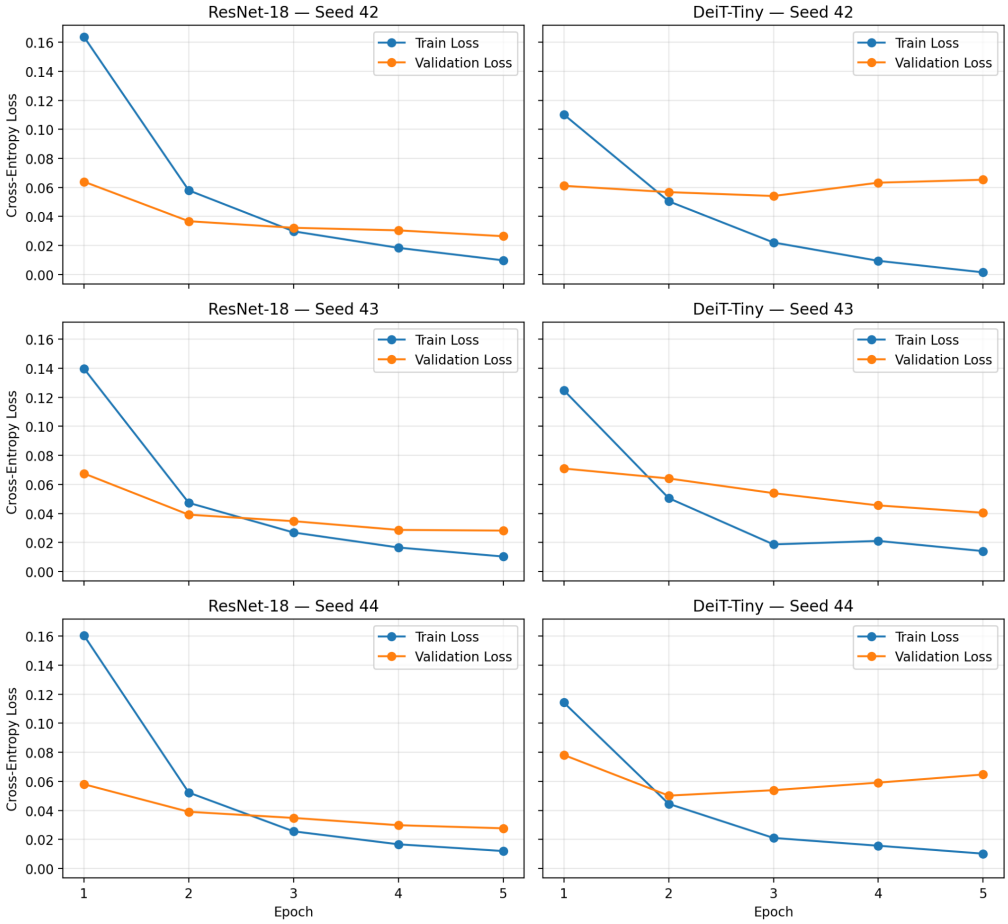


Figure 3: Training and validation cross-entropy loss over five epochs for each seed (rows) and model (columns). All six histories were read from saved CSV files; axes are shared.

Figure 3 shows decreasing or stabilizing validation loss for ResNet-18 in all three runs. DeiT-Tiny's validation loss rises in later epochs for seeds 42 and 44 even as training loss falls, suggesting possible overfitting in those runs. Five epochs and three seeds are insufficient to determine long-term convergence or the cause of these differences. Importantly, checkpoints were selected on *validation macro-F1*, not on minimum validation loss.

4.4 Classification errors

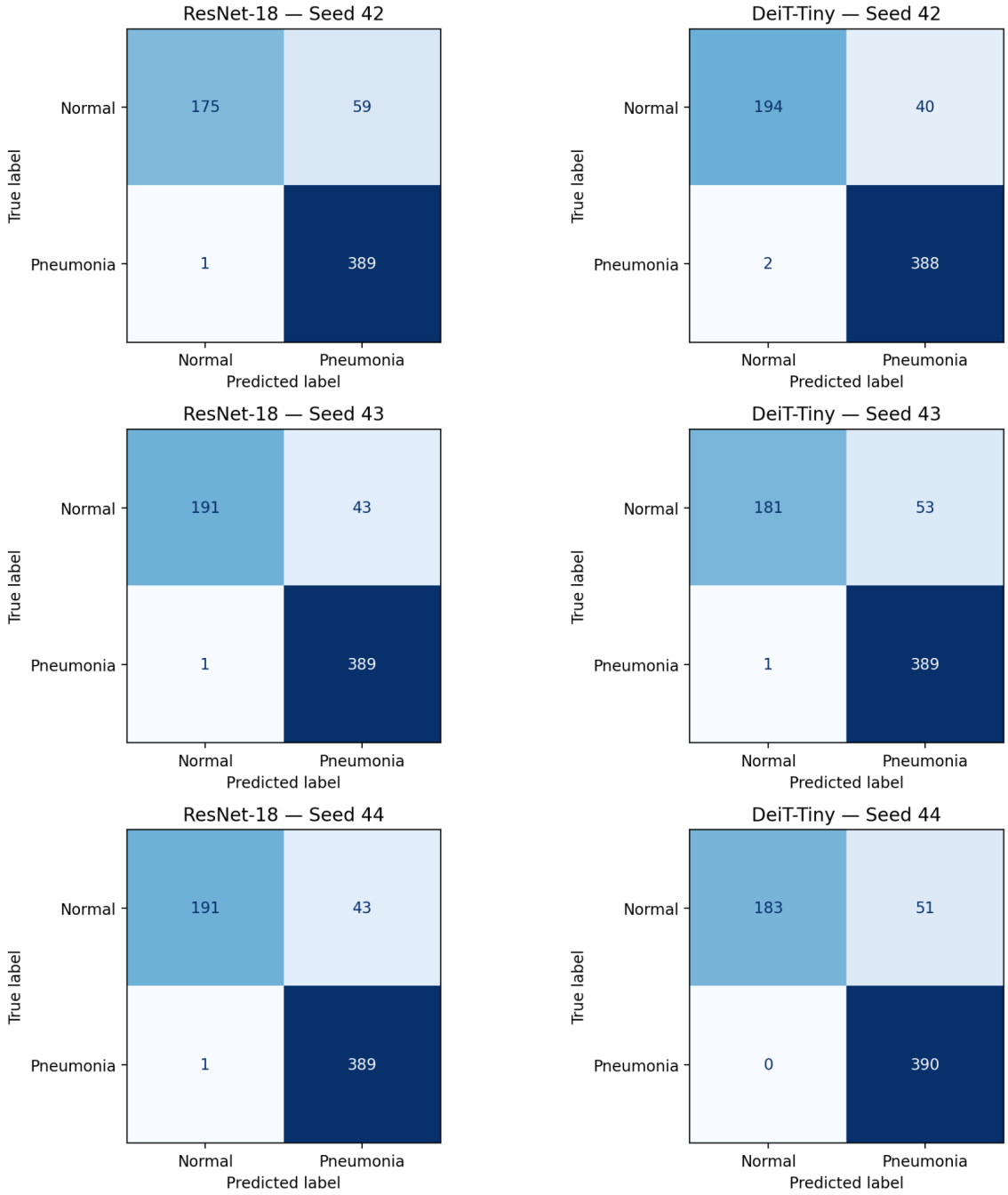


Figure 4: Six held-out test confusion matrices (234 Normal and 390 Pneumonia cases each). Rows correspond to seeds 42, 43, and 44; columns compare ResNet-18 and DeiT-Tiny. All displayed values are counts.

Across runs, there were only 0–2 false negatives, but 40–59 false positives. Consequently, mean sensitivity was 0.9974 for both models, whereas mean specificity was around 0.79. DeiT-Tiny had fewer false positives for seed 42; ResNet-18 had fewer for seeds 43 and 44. These rates describe errors at the fixed predicted-class rule (argmax), and do not constitute a clinical operating-point evaluation.

5 Discussion and Limitations

The validation-to-test macro-F1 drop was substantial for both model pipelines despite strong validation results. Pneumonia prevalence changed from approximately 74.2% in training/validation to 62.5% in test (Table 1), and the data partitions have different source roles. Such differences could contribute to observed performance gaps, but the present experiment cannot identify their causes. No adjustment or re-selection was made based on test performance.

The models differed not only in convolution versus self-attention, but also in parameter count and pretrained checkpoint/recipe. A shared training setup creates a consistent experimental comparison, yet does not isolate an architecture-only causal effect or ensure either architecture is individually optimized. Likewise, similar training runtimes on the recorded device do not establish similar inference cost, memory demand, or performance on other accelerators.

Study limitations. (1) The official test set contains only 624 images; no confidence intervals or significance/equivalence tests were performed. (2) Split proportions and provenance may affect generalization. (3) A fixed five-epoch budget and shared hyperparameters were used without independent model-specific tuning. (4) Architecture, model capacity, and pretrained weights vary together. (5) Three seeds provide only a limited assessment of training variability on one fixed test set. (6) There is no independent external-hospital evaluation. Accordingly, clinical reliability and broader generalizability are not established.

6 Conclusion

This controlled study compared ImageNet-pretrained ResNet-18 and DeiT-Tiny after full fine-tuning on PneumoniaMNIST+ 224. Their mean held-out test macro-F1 scores were 0.9114 and 0.9121 across three training seeds, with no consistent winner. Both models detected nearly all Pneumonia cases at the evaluated decision rule, but showed lower specificity and relatively frequent false positives for Normal images. ResNet-18 had more consistent validation loss trends, whereas some DeiT-Tiny runs showed late validation-loss increases. Mean training runtimes were similar. These findings support comparable performance for the two *specific pretrained pipelines* under the fixed protocol, not general CNN–Transformer equivalence. The limited test set, small number of seeds, common hyperparameters, and absence of external validation prevent clinical conclusions.

References

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Appendix A. Model and Experimental Pipeline Overview

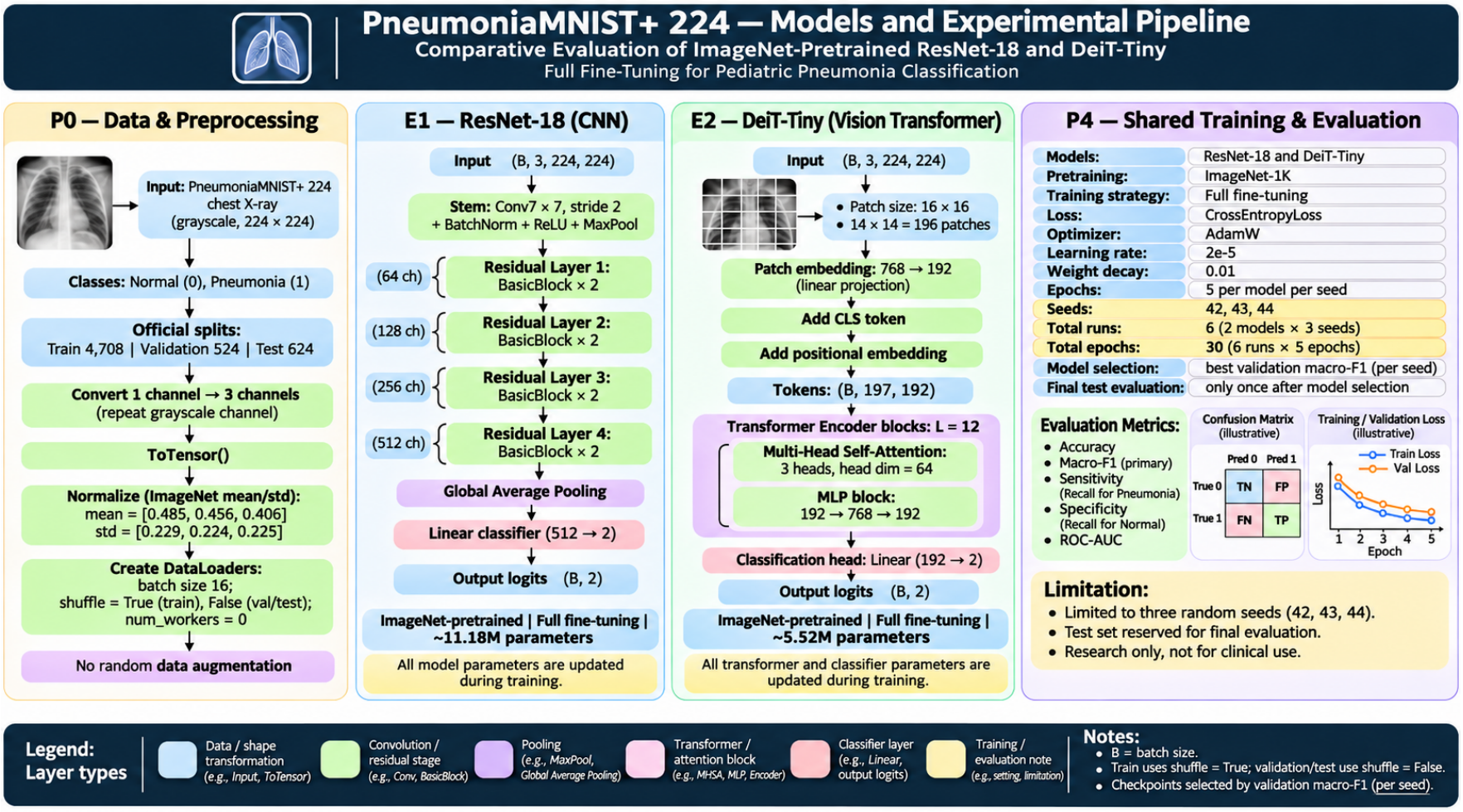


Figure A1. Overview of the experimental pipeline, including PneumoniaMNIST+ 224 preprocessing, ImageNet-pretrained ResNet-18 and DeiT-Tiny architectures, and the shared full fine-tuning and evaluation protocol across three random seeds.